

■ B.4 Evaluation

■ B.4.1 The Standard Evaluation Sequence

The following is the sequence of steps that *Mathematica* follows in evaluating an expression like $h[e_1, e_2, \dots]$. Every time the expression changes, *Mathematica* effectively starts the evaluation sequence over again.

- If the expression is a special object (e.g. `Integer`, `String`, etc.), leave it unchanged.
- Evaluate the head h of the expression.
- Evaluate each element e_i of the expression in turn. If h is a symbol with attributes `HoldFirst`, `HoldRest` or `HoldAll`, then skip evaluation of certain elements.
- If h has attribute `Flat`, then flatten out all nested expressions with head h .
- If h has attribute `Orderless`, then sort the e_i into order.
- If h has attribute `Listable`, then thread through any e_i that are lists.
- Use any applicable transformation rules that you have defined for objects of the form $f[h[e_1, \dots], \dots]$. (Also use transformation rules in which h appears in the head of an expression.)
- Use any built-in transformation rules for objects of the form $f[h[e_1, \dots], \dots]$.
- Use any applicable transformation rules that you have defined for $h[e_1, e_2, \dots]$.
- Use any built-in transformation rules for $h[e_1, e_2, \dots]$.